Jay Jungmin Ha

Address:
306 - 11 King's College Road, Toronto, Canada

Email: jay.ha@mail.utoronto.ca
Phone: (+1) 416-909-5115
Website: jayjmha.github.io

Education

University of Toronto, Toronto, Canada Sep. 2025 - Present
MASc Candidate in Electrical and Computer Engineering
Cumulative GPA: 4.0/4.0

Hanyang University, Seoul, Korea Mar. 2018 – Feb. 2025
B.S. in Electronic Engineering
Undergraduate Thesis: *Improving Training Methods to Enhance the Acceptance Rate of a Draft Model on EAGLE*
Cumulative GPA: 3.82/4.0 or 4.23/4.5

Research Experience

LLM Inference Acceleration Sep. 2025 – Present
Supervisor: Dr. Andreas Moshovos

- Research interest: Efficient Decoding, Speculative Decoding, Self-Speculative Decoding, Scheduling, Quantization, SW-HW co-optimization

LLM Inference Optimization Jul. 2024 – Dec. 2024
Supervisor: Dr. Jungwook Choi

- Undergraduate Researcher (AIHA Lab: Artificial Intelligence Hardware & Algorithms Lab)
- Worked on optimizing LLM inference through various methods such as speculative decoding, vLLM, and FlashAttention to address bottlenecks in autoregressive LLMs
- Implemented Speculative Decoding Algorithm on a neural processing unit (ATOM NPU) from Rebellions

Samsung Computer Engineering Challenge 2024 Aug. 2024 - Sep. 2024
Supervisor: Dr. Jungwook Choi

- Collaborated in an AIHA Lab team to optimize LLM inference on the edge device (Jetson AGX Orin 32GB), under competition constraints that prohibit model modifications (i.e., no pruning or quantization) while focusing on system-level optimization
- Applied speculative decoding and GPU kernel optimization to reduce inference latency without impacting model accuracy

Current-Mode & Voltage-Mode Stimulator Circuit Design Nov. 2023 - Jan. 2024
Supervisor: Dr. Jeonghee Kim

- Undergraduate Researcher (Hanyang Bio-signal Processing and NeuroEngineering Lab)
- Conducted a preliminary review of existing research and developed a robust sensory electrical stimulator resilient to skin impedance variations while utilizing constant current and voltage sources

Research Conferences

- **Ha J., Ganesan K., Nguyen A., Ahmed T., Moshovos A. (2026) Characterizing self-speculative decoding approaches for accelerating LLMs. AdatFM: Resource-Adaptive Foundation Model Inference (ICML2026 Workshop).**
- **Ha J.**, Choi JH., and Choi JW. (2024) *Improving Training Methods to Enhance the Acceptance Rate of a Draft Model on EAGLE.* Poster presented at the 2024 Korean Artificial Intelligence Association Conference, Samsung SDS, Seoul, South Korea. Nov 22, 2024.

Honors and Awards

- Percy Edward Hart Award ($13,020) 2026
- 2024 HYU Systems & Semiconductors Co-op Track Scholarship ($1,000) 2024
- Hanyang University Academic Excellence Scholarship ('Hanyang Brain') ($4,630) 2023
- 2023 HYU Systems & Semiconductors Co-op Track Scholarship ($2,800) 2023
- HYU merit-based scholarship (given to students with academic excellence) ($200) 2023
- Hanyang Academic Excellence Award – Top 1% ranked in University 2023
- Academic Excellence Award, Spring–awarded to students with GPA>4.0/4.5 2018, 2019, 2023, 2024
- Academic Excellence Award, Fall–awarded to students with GPA > 4.0/4.5 2018, 2019, 2023

Academic Projects

RISC-V Based SoC Design and Implementation Sep. 2024 – Dec. 2024
Hanyang University *ELE 4105 (Capstone Design: Digital Circuits)*

- Designing a complete system with a RISC-V processor (SweRV), integrating BUS and peripherals, and deploying the design on an FPGA

Assistance System for the Blind using Object Detection (YOLOv7) Mar. 2024 - Jun. 2024
Hanyang University *ELE 4103 (Intro to AI Platform and Lab)*

- Designed and implemented a real-time pedestrian assistance system for visually impaired individuals, utilizing Jetson Nano board
- Fine-tuned YOLOv7 model for object detection with a custom dataset, optimized it with TensorRT for real-time, on-device performance, and developed an auditory guidance system incorporating an object tracking algorithm

CNN Hardware Implementation on FPGA Mar. 2024 - Jun. 2024
Hanyang University *ITE 4003 (SoC Design)*

- Implemented hardware for CNN operations on a DE1-SoC board, addressing FPGA resource limitations by designing pipelined CNN cores
- Developed memory access mechanism using on-chip SRAM blocks for communication between FPGA and HPS

32-Bit 5-Stage Pipelined RISC-V Processor Mar. 2023 - Jun. 2023
Hanyang University *ENE 1004 (Computer Architecture)*

- Implemented a 32-bit 5-stage RISC-V processor with a 2-bit branch predictor using Verilog
- Optimized the Fibonacci algorithm in Assembly Language and demonstrated its performance on FPGA

Teaching Experience

Teaching Assistant (ECE253 Digital and Computer Systems) - University of Toronto Sep. 2026 - Present
Teaching Assistant (ECE244 Programming Fundamentals) - University of Toronto Sep. 2026 - Present
Teaching Assistant (ECE243 Computer Organization) - University of Toronto Jan. 2026 - Apr. 2026

International Engineering Student Tutoring Program - Hanyang University Apr. 2024 - Jun. 2024

- Tutored international engineering students in Calculus 1, focusing on fundamental concepts and problem-solving techniques

Mathematics Tutor - PROCESS Mathematics Academy Feb. 2020 - May. 2020

- Taught high school students Calculus, Probability and Statistics, and Geometry and Vectors in preparation of university entrance exams in Korea

Volunteer and Service Experience

International Student Mentor - Hanyang Institute of International Education Mar. 2023 - Jun. 2023

- Assisted international students adapt to Korean culture and language through team building activities

Social Service Personnel – Eoulrim Sheltered Workshop Oct. 2020 - Jul. 2022

- Fulfilled mandatory military service by working in a sheltered workshop for individuals with disabilities, assisting in vocational training programs, and providing support in daily work activities

Skills

Relevant Coursework

- ECE1755 Parallel Computer Architecture and Programming
- ECE1513 Introduction to Machine Learning
- ECE552 Computer Architecture
- ECE1718 Embedded Systems with Linux
- ECE1756 Reconfigurable Computing and FPGA Architecture
- ELE 4105 Capstone Design: Digital Circuit
- ITE 4003 SoC Design
- ELE 4103 Intro to AI Platform
- APA 4055 Introduction to AI
- ENE 1004 Computer Architecture
- ELE 3021 Operating Systems

Programming Languages: Python, C/C++, Assembly Language, LaTeX
Tools: NVIDIA Nsight Systems, PyTorch, NumPy, Pandas